

# TEK4090 – Introduction to Modern Control

## Assignment 2 v4.0

Due: October 23 2024, 13:15

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1. (10 points) Consider the continuous-time system,

$$\ddot{x} = u \tag{1}$$

$$J(t_0) = \frac{1}{2}x^2(0) + r \int_0^2 u^2 dt. \tag{2}$$

Hint: read §6.3 in Lewis, Syrmos, & Vraibie for some relevant example problems

- (a) Write the Hamilton-Jacobi-Bellman (HJB equation), eliminating  $u(t)$

**Ans:** solution here

- (b) Assume that the optimal cost-to-go function  $J^*(t)$  is given by,

$$J^*(t) = \frac{1}{2}s_1(t)x^2(t) + s_2(t)x(t)\dot{x}(t) + \frac{1}{2}s_3(t)\dot{x}^2(t). \tag{3}$$

Use the HJB equation to derive coupled differential equations for these  $s_i(t)$ 's, and find boundary conditions for these equations.

- (c) Write the linear state feedback optimal control policy in terms of the  $s_i(t)$ 's.

**Ans:**

(a)

2. (10 points) Pick a system from a paper that you have found on a topic that you are interested in, potentially the topic you will choose for your final project.

- (a) What is/are the paper(s) you are drawing from?
- (b) What are the dynamics of the system in question? What are the inputs and outputs?
- (c) What is the system trying to achieve?
- (d) What are the challenges for the system?
- (e) What control tools can help the system overcome the challenges in achieving its goals?

3. (20 points) Suppose that  $f : \mathbb{R} \mapsto \mathbb{R}$  is convex, and  $a < b$  are in the domain of  $f$ .

(a) Show that

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) \quad (4)$$

for all  $x \in [a, b]$

(b) Show that

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x} \quad (5)$$

for all  $x \in (a, b)$ . Draw a sketch that illustrates this inequality

(c) Suppose that  $f$  is differentiable. Use the result in (b) to show that

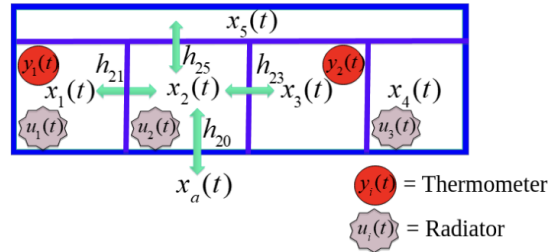
$$f'(a) \leq \frac{f(b) - f(a)}{b - a} \leq f'(b). \quad (6)$$

(d) Suppose  $f$  is twice differentiable. Use the result in (c) to show that  $f''(a) \geq 0$  and  $f''(b) \geq 0$ .

**Ans:**

(a)

4. (20 points) (LQR for Buildings) Consider a multi-zone building as in Fig. 2.



**Fig. 2:** Building model schematic

- (a) Write down the dynamical equations of the building in state-space form, including a term for disturbances  $d(t)$ , and for the ambient temperature  $x_a(t)$ , assuming the parameters are

$$\frac{h_{ij}}{C_i} = \begin{cases} 10 \text{ W/}^\circ\text{K} & i \text{ is adjacent to } j \\ 0 & \text{else} \end{cases} \quad (7)$$

$$C_i = 1.08 \times 10^6 \text{ J/}^\circ\text{K} \quad (8)$$

- (b) Consider an LQR cost function of the form

$$\int_0^T [(x - \bar{x})^T Q (x - \bar{x}) + u^T R u] dt, \quad (9)$$

where  $T = 24\text{h} \times 60\frac{\text{m}}{\text{h}} \times 60\frac{\text{s}}{\text{m}} = 86400\text{s}$  is one day in seconds. Set  $\bar{x} = 20^\circ$ , and write the dynamics in the coordinates  $z := x - \bar{x}$ .

- (c) Note that driving the variable  $z(t) \rightarrow 0$  is therefore equivalent to driving  $x(t) \rightarrow 20^\circ$ . Write a `matlab` script to solve the finite-time horizon LQR problem. Select a positive semi-definite  $Q$  and a positive-definite  $R$  that gives you “decent” results. Assume the initial conditions, disturbance, and ambient temperature follow:

$$x_i(0) = 15 - i \times \text{sign}\{(-1)^i\} \quad [^\circ \text{C}] \quad (10)$$

$$d(t) = \max \left[ 0, 500 \sin \left( \frac{2\pi t}{86400\text{s}} \right) - 300 \right] \quad [\text{W}] \quad (11)$$

$$x_a(t) = 20^\circ + 5 \sin \left( \frac{2\pi t}{86400\text{s}} \right) \quad [^\circ \text{C}] \quad (12)$$

where the sign function is:

$$\text{sign}(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \\ 0 & x = 0. \end{cases} \quad (13)$$

**Note:** If solving the finite-time problem proves too difficult, feel free to solve the infinite-time problem instead. `Matlab` has a built-in function called `lqr` that produces the (constant) gain  $K$ ; read the documentation for details.

- (d) Provide:

- A description of your simulation code
- Plots of  $d(t)$ ,  $x_a(t)$ ,  $x_i(t)$ , and  $u(t)$ .
- A discussion of the effects of varying  $Q$  and  $R$

**Ans:**

(a)

5. (10 points) Suppose that  $Q = \mathbf{0}$ , the zero matrix, and  $R = I$ . Prove that

$$x_0^T S_0 x_0 = \min_{\{u_k\}_{k=0}^{T-1}} \sum_{k=0}^{T-1} \|u_k\|_2^2, \quad (14)$$

meaning that  $J_0 = x_0^T S_0 x_0$  is the minimum control energy required to drive the system to zero.

**Ans:**

6. (10 points) *Crassidis & Junkins* Problem 3.14

**Ans:**

7. (10 points) *Crassidis & Junkins* Problem 3.15

**Ans:**

8. (10 points) *Crassidis & Junkins* Problem 7.4

**Ans:**